

PUNEETH JOSEPH

Principal Software Engineer

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PROFESSIONAL SUMMARY

Principal Software Engineer with 14+ years of experience specializing in automotive operating systems and ADAS Domain Controller development. Led development and integration of MB.OS (Mercedes-Benz Operating System), a Linux and QNX-based platform enabling software-driven architecture. Expert in software-driven vehicle architecture integration, Adaptive AUTOSAR, QNX, Linux, and embedded systems architecture. Proven track record at Mercedes-Benz Group and Robert Bosch, delivering next-generation software-driven automotive platforms integrating multiple sensor technologies (Cameras, Radar, Lidar, Ultrasonic). Strong background in software architecture, system integration, and cross-functional collaboration with platform teams, application developers and external partners.

TECHNICAL SKILLS

Automotive Software Architecture

- MB.OS (Mercedes-Benz Operating System)
- Software-Driven Architecture
- Adaptive AUTOSAR, AUTOSAR Classic
- QNX, Linux, POSIX Systems
- Service-Oriented Architecture (SOA)
- Embedded Software Architecture

Programming & Scripting

- C++, C
- Shell Scripting
- Python (automation & tooling)

Development Tools & Platforms

- QNX Momentics IDE
- Nvidia Drive Platform (Orin SoC)
- Debugging Tools (GDB, JTAG)
- CI/CD Tools
- Git, SVN
- Hypervisor Technologies

Domain Expertise

- MB.OS Development & Integration
- ADAS Domain Controllers
- ECU Development & Integration
- Virtual ECU (vECU) Development
- Image Processing (CNN, SSD)
- Sensor Integration
- Navigation Systems Middleware
- Over-the-Air (OTA) Updates
- Level 3 Autonomous Driving Systems

PROFESSIONAL EXPERIENCE

Principal Software Integrator

July 2021 - October 2025

Mercedes-Benz Group

Stuttgart, Germany

Lead Software Integrator for Mercedes-Benz Operating System (MB.OS) development and ADAS Domain Controller projects, delivering next-generation software-driven architecture and platform solutions integrating Linux, QNX, AUTOSAR Classic, and AUTOSAR Adaptive systems.

- Developed and integrated MB.OS (Mercedes-Benz Operating System), a Linux and QNX-based platform for the Mercedes Modular Architecture (MMA), enabling software-driven architecture with over-the-air updates and full software stack control
- Developed and integrated multi-sensor ADAS platform supporting Cameras, Ultrasonic sensors, Radar, and Lidar technologies. Integrated dual-OS architecture (QNX for production safety-critical systems and Linux for development) and Nvidia Drive platform with Orin system-on-chip (SoC) for advanced computing capabilities, supporting Level 3 autonomous driving features
- Collaborated with platform teams, application developers, and external partners (including Nvidia) to define software architecture and integration strategies
- Maintained and released Bring-Up Software for Release Boards, enabling pre-delivery evaluation and validation of release software
- Served as Technical Contact Partner for internal and external customers, providing expertise on software integration and troubleshooting
- Customized QNX OS for custom ECU hardware from Nvidia and Qualcomm using QNX Momentics IDE
- Designed embedded software architecture for microprocessor-based automotive systems
- Implemented Service-Oriented Architecture solutions for embedded domain

Specialist Software Engineer

August 2013 - June 2021

Robert Bosch Engineering and Business Solutions

Bengaluru, India

Developed ADAS software components and middleware solutions for automotive ECUs, working on cutting-edge projects including Virtual ECU development and AI-based object detection systems.

- Developed virtual ECU (vECU) software solutions for multiple OEMs, enabling simulation and testing of real ECU functionality
- Designed and optimized Adaptive AUTOSAR-based applications for real-time object detection, processing frame data from camera sensors
- Deployed deep learning algorithms (CNN and SSD) for object detection and classification on High Performance Computing platforms
- Contributed to middleware development for navigation systems, implementing both basic and advanced navigation features
- Architected Proof of Concept (POC) solution integrating Android, Telematics (QNX), and Bosch Navigation System (Linux) using Hypervisor technology

Software Developer

August 2010 - July 2013

Wipro Technologies

Bengaluru, India

Led platform migration project, porting EVDO application software from Solaris to Linux-based hardware platform.

- Successfully ported C++ application codebase and shell scripts from Solaris to Linux platform
- Implemented Linux-specific wrappers for shell commands, ensuring seamless functionality across platforms
- Resolved endianness compatibility issues in C++ codebase for cross-platform support
- Enhanced application features during migration process

PERSONAL PROJECTS

ConTra - Contracts Tracker

iOS & macOS App

Contract and subscription management platform for macOS and iOS. Designed with intuitive user experience and robust security architecture, the platform simplifies financial tracking and management of contracts, subscriptions, rental agreements, and insurance policies. Built on a privacy-first foundation with advanced analytics and comprehensive reporting capabilities.

Technologies: Swift, SwiftUI, macOS, iOS, Core Data

EDUCATION

Bachelor of Engineering in Information Science

Visvesvaraya Technological University

Bengaluru, India | July 2007 - June 2010

CERTIFICATIONS

Certified Scrum Product Owner® (CSPO®)

Scrum Alliance

December 2023

ADDITIONAL INFORMATION

Languages: English (Fluent), German (Working Proficiency), Hindi (Native)

Interests: Reading, Performance Monitoring, Football, Technology Research